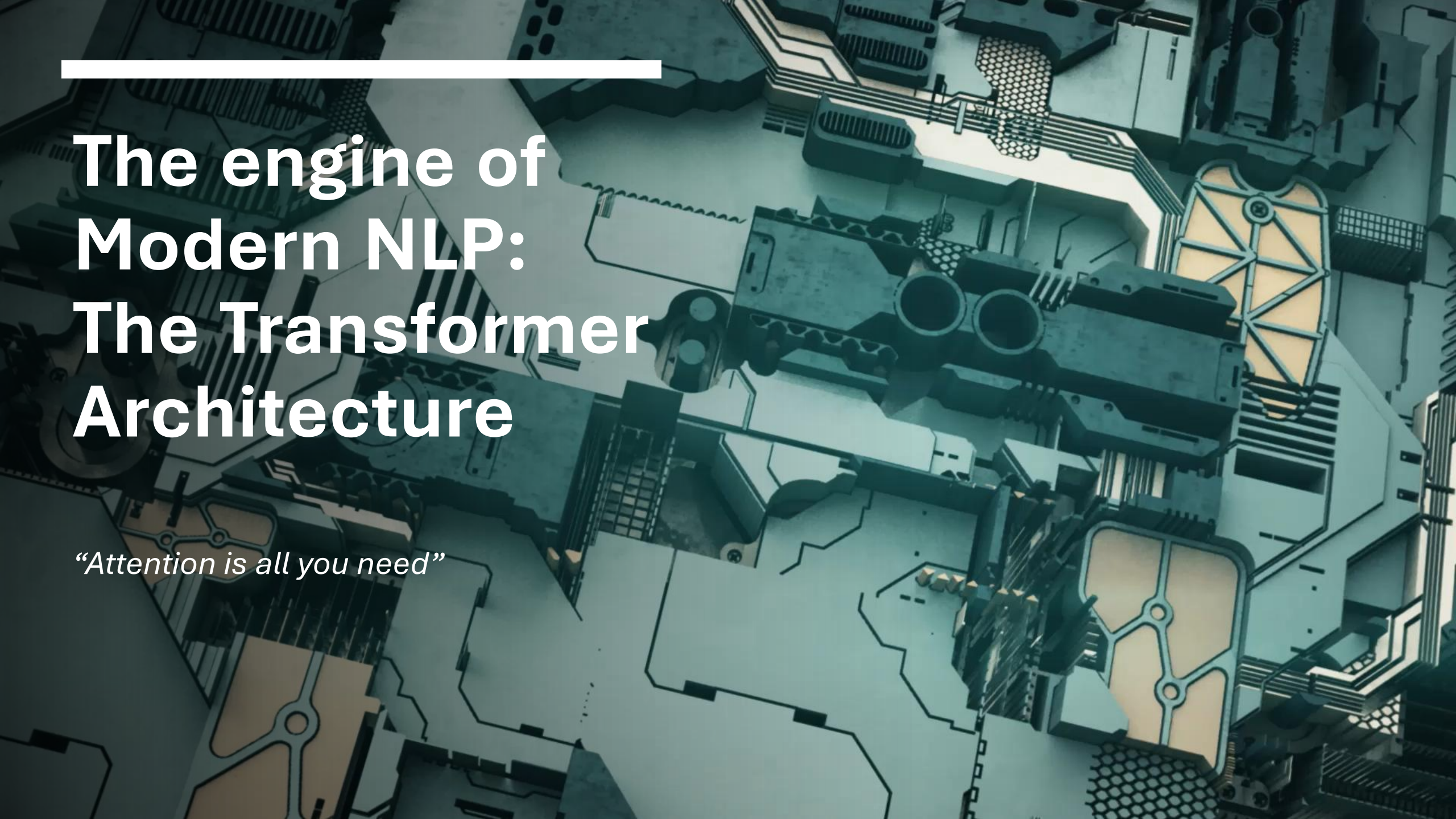




The engine of Modern NLP: The Transformer Architecture

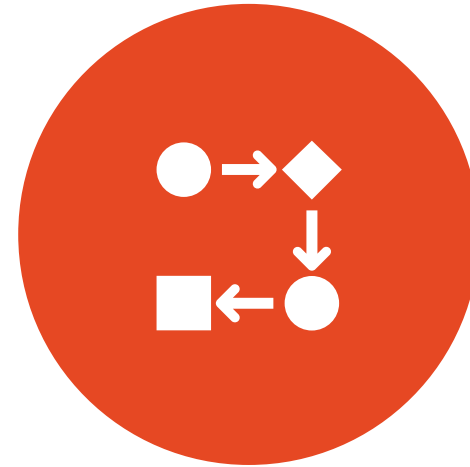
“Attention is all you need”

The Paradigm Shift

The Engine of Modern AI: The Transformer Architecture



[HTTPS://ARXIV.ORG/ABS/1706.03762](https://arxiv.org/abs/1706.03762)



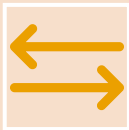
FROM SEQUENTIAL RNNS TO
"ATTENTION IS ALL YOU NEED"

The Bottleneck of RNNs (The "Straw" Problem)



The Problem: Recurrent Neural Networks (RNNs/LSTMs) process data sequentially.

Analogy: It's like reading a book through a drinking straw. You only see one word at a time, and by the time you reach the end of a long sentence, you've forgotten the beginning.



The Vanishing Gradient: Mathematically, the signal from the first word gets weaker as it travels through time.



Imagine: A game of "Telephone" where the message gets distorted at the end of the line

The Solution: Parallelism & Attention

The Transformer Proposal (2017):

Parallelism: Process the *entire* sentence at the same time. No more steps.

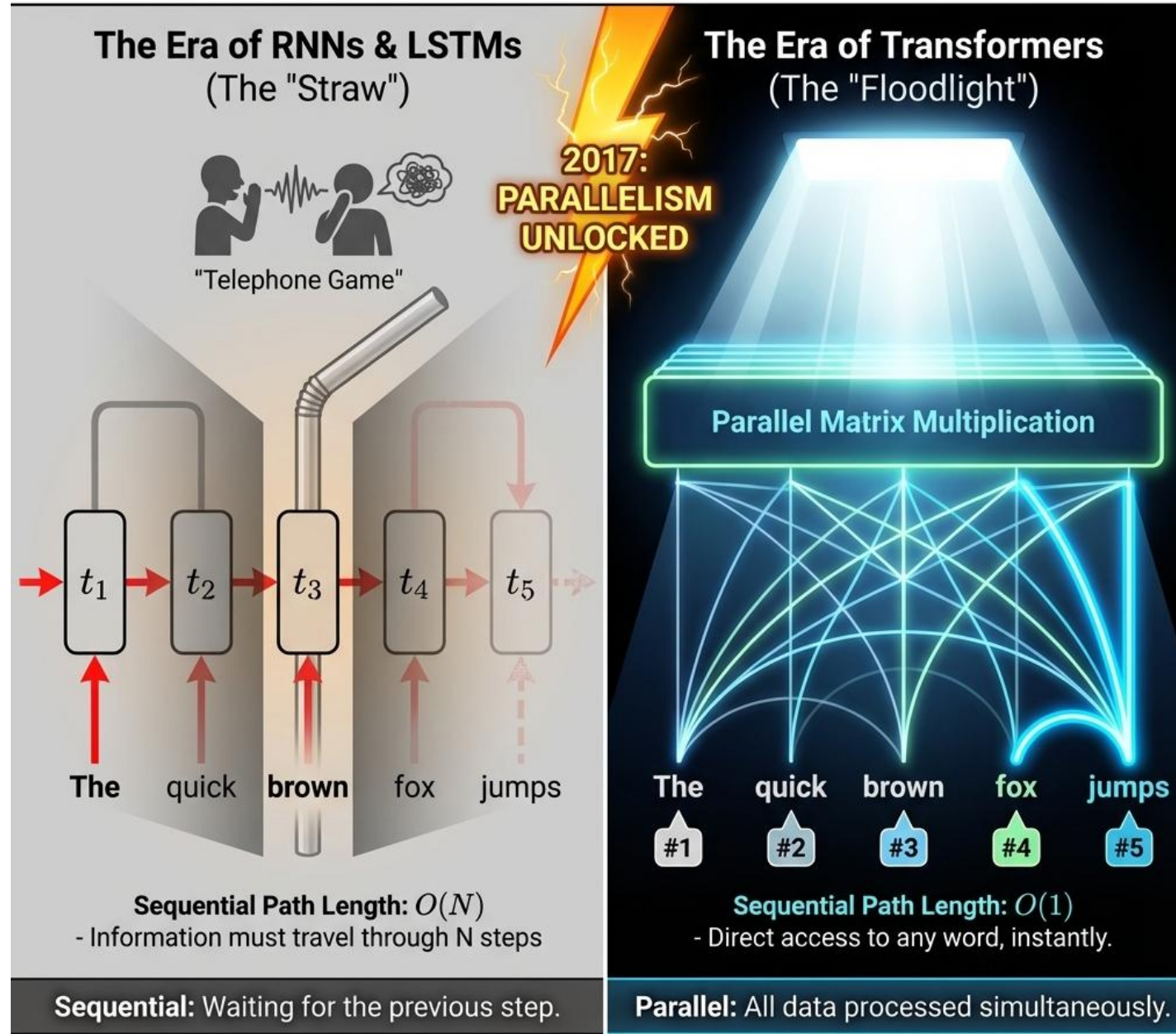
Attention: Instead of relying on memory, look at every other word directly to find context.



Impact: This allowed models to train on the entire internet (Wikipedia, Common Crawl) efficiently

The Paradigm Shift: From Sequential Chains to Parallel Webs

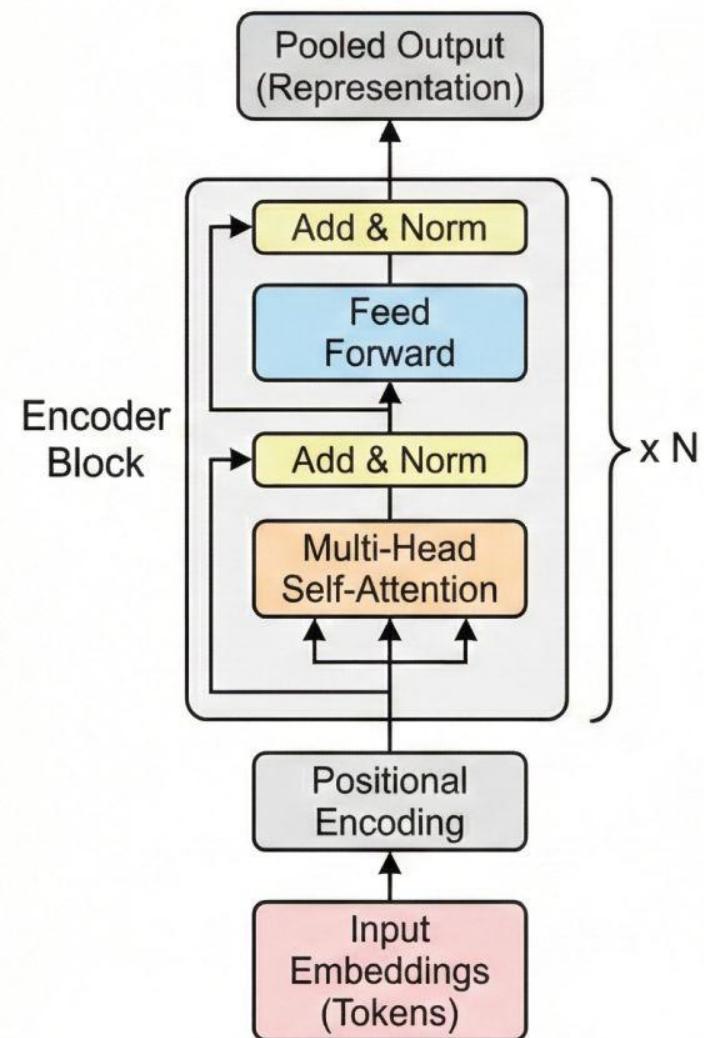
How Attention Collapsed Time into Space



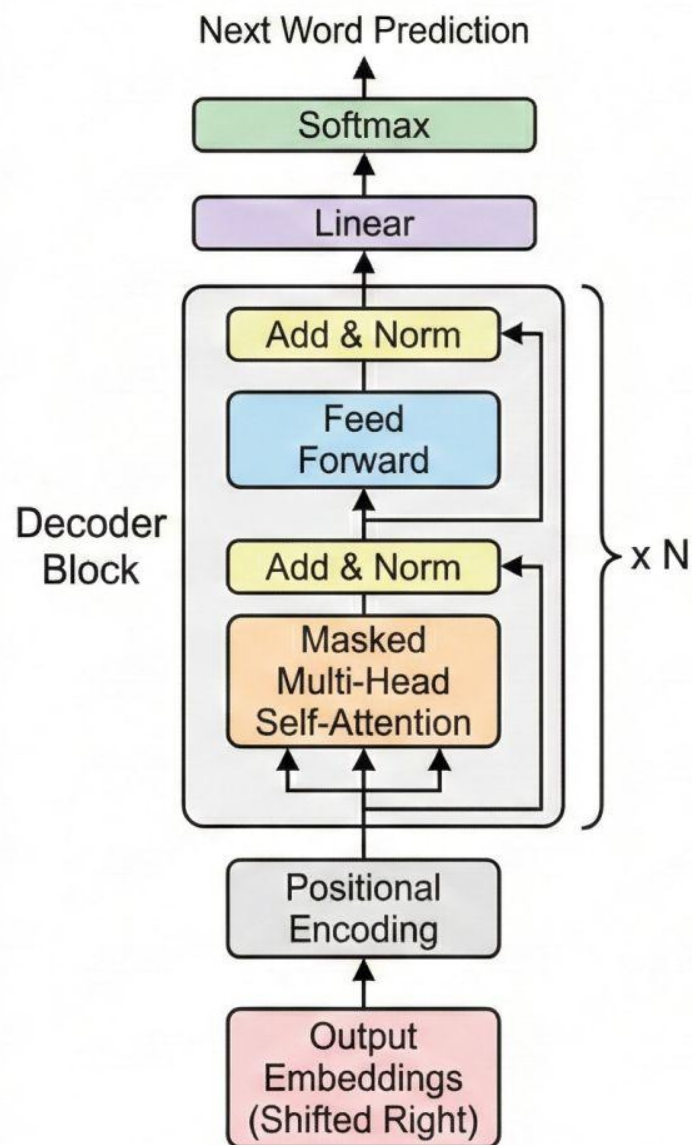
Architectures: Encoder, Decoder & Bidirectional Transformer

A white square electronic device with four screws is centered in the image. A white cable extends from the right side of the device. The background is dark gray with wavy lines on the left side. The title text is overlaid on the left side of the image.

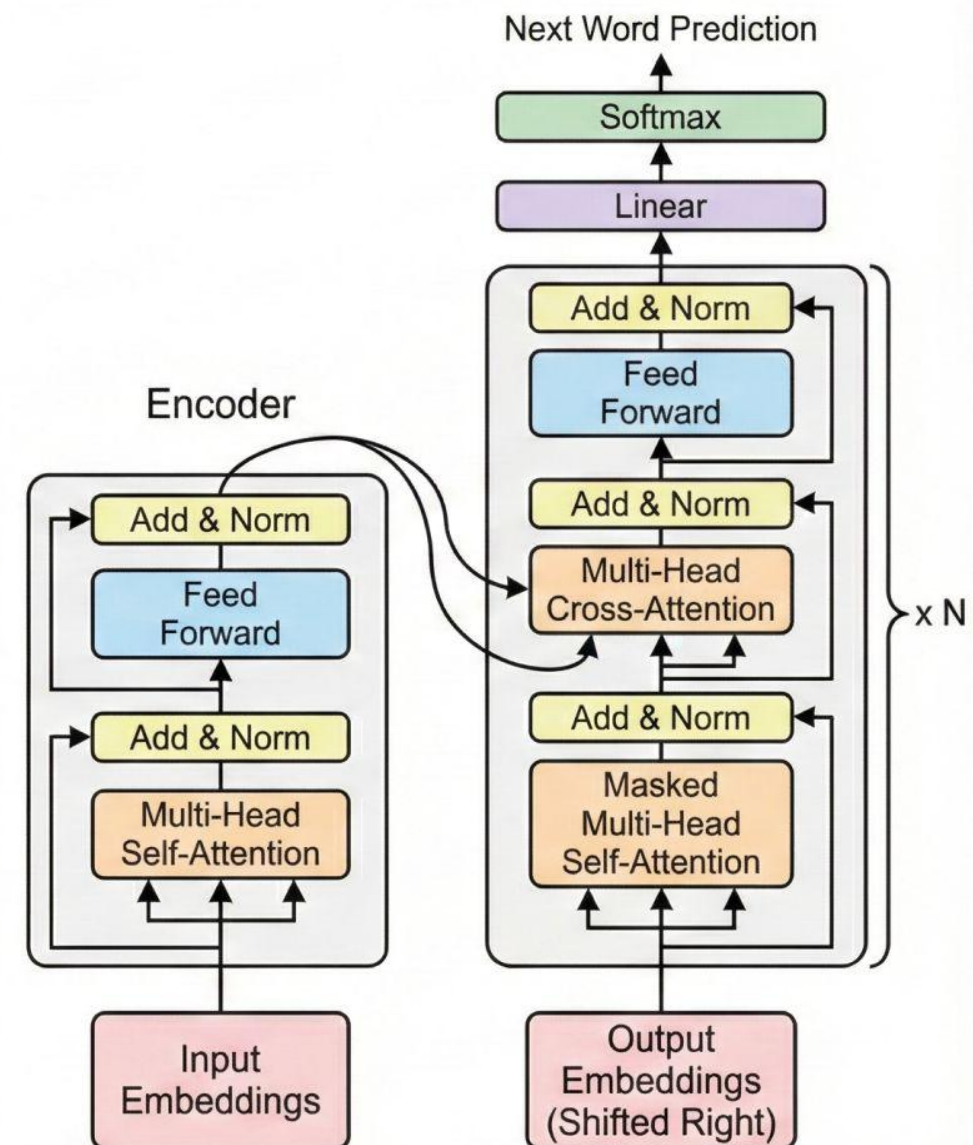
1. Encoder Architecture (e.g., BERT)



2. Decoder Architecture (e.g., GPT)



3. Bidirectional Original Transformer (e.g., Vaswani et al.)



Deep Dive into Self- Attention



Self-Attention: The Core Concept

- **Definition:** A mechanism that allows a word to look at other words in the sentence to gather context.
- **Example:** "The **animal** didn't cross the **street** because **it** was too tired."
 - *Question:* What does "**it**" refer to?
 - *Attention:* The model builds a strong link between "**it**" and "**animal**" and a weak link between "**it**" and "**street**."

The Database Analogy (Q, K, V)

This is the most critical concept to teach correctly.

Every word produces three vectors:

- **Query (Q):** What am I looking for? (*"I am a pronoun looking for the noun I replace."*)
- **Key (K):** What defines me? (*"I am a noun, specifically an animal."*)
- **Value (V):** What information do I hold? (*"I am the vector representation of 'Animal'."*)

The Mechanism:

- We compare the **Query** of word A with the **Keys** of every other word.
- High match = High Attention Score.

The "Cocktail Party" Analogy

Scenario: You are at a noisy party.

Q (Your Focus): You are listening for your name or a specific topic (e.g., "Robotics").

K (Voices): Everyone else speaking acts as a "Key".

Score: When someone says "Robotics" (High match between Q and K), you "attend" to them.

V (Content): You absorb the information they are saying.

Self-Attention: The Engine of Transformers - Understanding the Core Concept

1 The Core Concept: Definition & Example.

Definition: A mechanism that allows a word to look at other words in the sentence to gather context.^[1]

The **animal** didn't cross the street because **it** was too tired.

Strong Link (Referent)

Weak Link

Question: What does "it" refer to?
Attention: The model builds a strong link between "it" and "animal" and a weak link between "it" and "street."^[1]

2 The Database Analogy: Query, Key, Value (Q, K, V).



Query (Q)
What am I looking for?
(I am a pronoun looking for the noun I replace.)

The Mechanism:
We compare the Query of word A with the Keys of every other word.
High match = High Attention Score.^[1]

Value (V)
What information do I hold?
(I am the vector representation of 'Animal'.)

3 The "Cocktail Party" Analogy.

You are listening for your name or a specific topic (e.g., "Robotics").

Weather...
Sports...
Stock Market...

Robotics



Q (Your Focus)
When someone says "Robotics" (High match between Q and K), you "attend" to them.

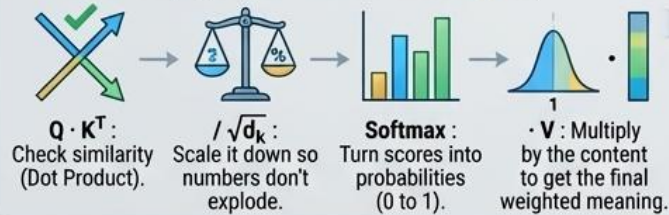
High Match (Score)

V (Content)
You absorb the information they are saying.

4 The Math of Attention (Simplified).

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q} \cdot \mathbf{K}^T}{\sqrt{d_k}}\right) \cdot \mathbf{V} \text{ [2]}$$

Step-by-step visual translation to $\mathbf{Q} \cdot \mathbf{K}^T$:



5 Multi-Head Attention (The "Committee").

Why one head isn't enough: Relationships are complex.

Use 8 or 12 "Heads" in parallel.



Visual: A committee where one person takes notes on dates, another on names, another on action items...

Head 1: Focuses on grammar (Subject-Verb).

Head 2: Focuses on vocabulary links (King-Queen).

Head 3: Focuses on long-distance dependencies.



The Architecture Family

The Transformer Block (The Lego Brick)

- **Anatomy:**
 - **Self-Attention:** Finds context.
 - **Layer Norm:** Keeps numbers stable (crucial for deep networks).
 - **Feed Forward Network:** Processes the information.
 - **Residual Connection:** A "highway" that lets data skip layers (prevents signal loss).

Positional Encodings (The Time Stamp)

- **The Problem:** Since transformers look at everything at once, they don't know that "Man bites Dog" is different from "Dog bites Man."
- **The Fix:** We add a "timestamp" vector to every word so the model knows its order in the sequence.

The Family Tree: Encoders vs. Decoders

- **Encoder-Only (BERT):**

- *Visual:* Eyes looking both Left and Right (Bidirectional).
- **Use:** Understanding, Classification, NER. "Fill in the blank."

- **Decoder-Only (GPT):**

- *Visual:* Blinders on the right eye. Can only see what came before (Unidirectional).
- **Use:** Generation, Chat, Writing code. "Predict the next word."

- **Encoder-Decoder (T5 & BART)**

- **The Hybrid:** Uses an Encoder to read the input (e.g., French) and a Decoder to write the output (e.g., English).
- **Use:** Translation, Summarization.

The Transformer Architecture: Core Blocks, Positional Encoding, and Family Tree

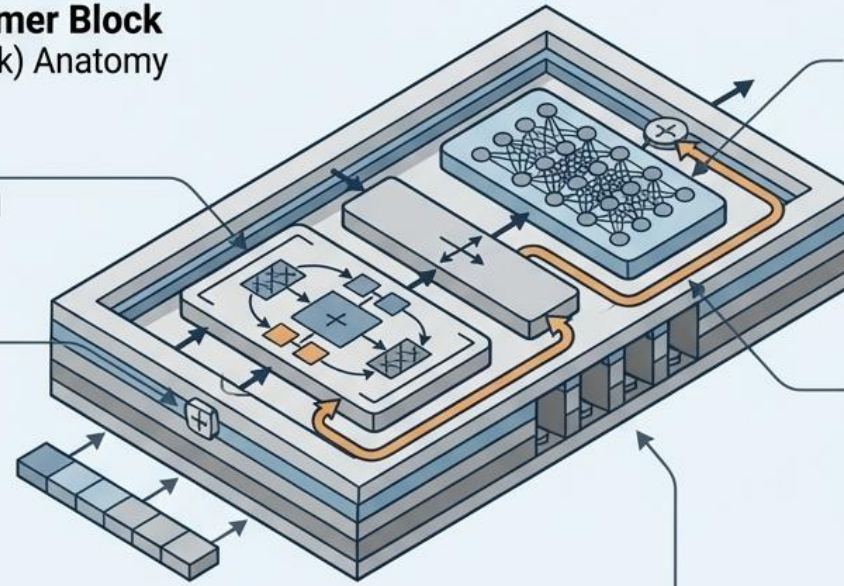
The Transformer Block (The Lego Brick) Anatomy

Self-Attention:
Finds context.[1][3]

Layer Norm:
Keeps numbers stable (crucial for deep networks).[4]

Feed Forward Network:
Processes the information.[1]

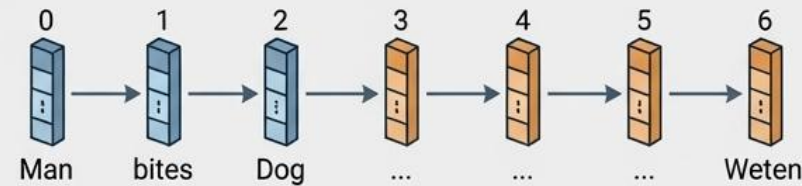
Residual Connection:
A "highway" that lets data skip layers (prevents signal loss)



Positional Encodings (The Time Stamp)

The Problem: Transformers are permutation invariant.

The Fix: Add timestamp vector for order.



The Family Tree: Encoders vs. Decoders

Encoder-Only (BERT)



[CLS] Fill in the [MASK] [SEP]

Bidirectional context.[5]

Use: Understanding, Classification, NER.[6]

Decoder-Only (GPT)



Predict the next [word]...

Unidirectional context.

Use: Generation, Chat, Writing code.[7]

Encoder-Decoder (T5 & BART)



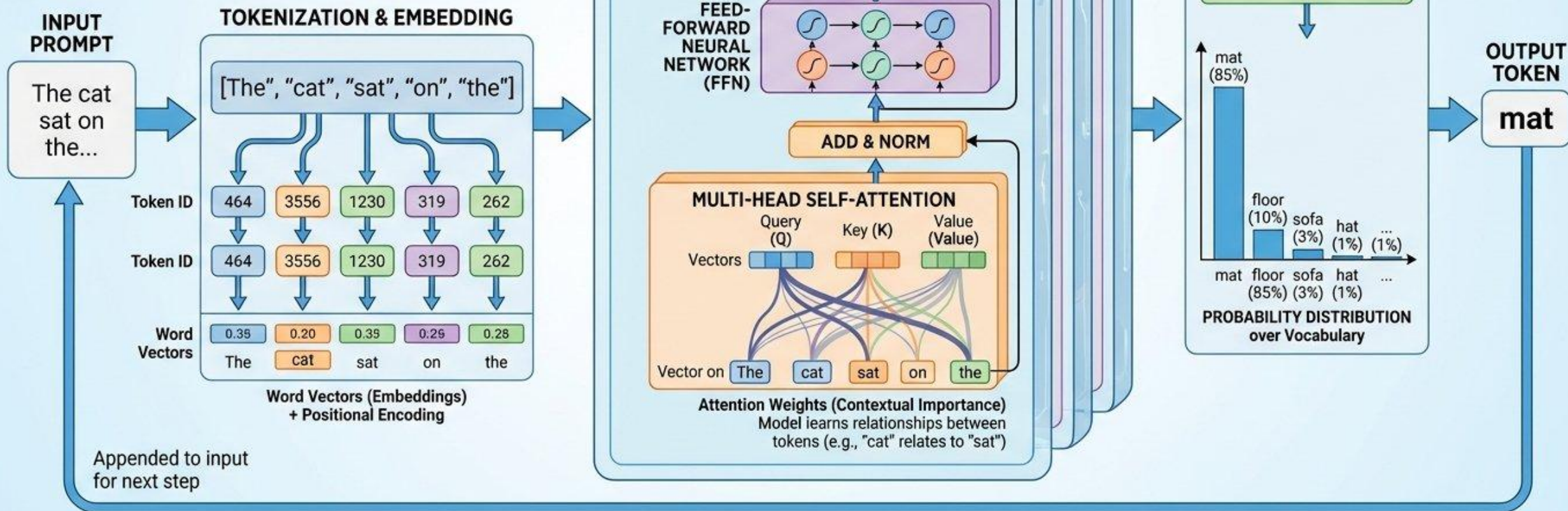
Bonjour le monde Hello the world

Hybrid: Encoder reads input, Decoder writes output.[1]

Use: Translation, Summarization

INSIDE THE TRANSFORMER LLM:

How It Generates Text



Recommended Readings:

- **The Gold Standard Visualization:** [The Illustrated Transformer by Jay Alamar](#) (Essential reading).
- **Interactive Code:** [The Annotated Transformer \(Harvard NLP\)](#) - A line-by-line PyTorch implementation.
- **Visualization Tool:** [BertViz GitHub Repository](#) - For the colab: <https://colab.research.google.com/drive/1hXIQ77A4TYS4y3UthWF-Ci7V7vVUoxmQ?usp=sharing>
- **Transformer Explainer:** <https://poloclub.github.io/transformer-explainer/>